Auto Loader is a powerful tool. It can be used standalone, in combination with **Spark Structured Streaming** (a near real-time processing engine that we'll return to in *Chapter 5*), or as the underlying technology for Delta Live Tables (which we'll introduce shortly). For an example of streaming ingestion, please refer to the Streaming Transactions project in *Applying our learning*.

Once you have set up your data ingestion process, it's important to build a resilient pipeline that helps you monitor your data quality. Let's discuss how Auto Loader's ability to evolve a schema over time plays a key role in data quality.

Schema Evolution

Auto Loader's schema evolution allows you to add or modify fields in streaming data seamlessly. Databricks automatically adjusts relevant data to fit the new schema while preserving existing data integrity. This automated schema handling makes it easy to evolve your data schema over time as your business needs and data sources change without worrying about data loss or downtime. Furthermore, schema handling reduces the amount of work and headache associated with managing data pipelines that could change unexpectedly.

In the Applying our learning section's Transaction Streaming project, we put Schema Evolution to work. You'll see how no data is lost despite the transaction data's schema changing without notice. In our case, the default schema evolution mode for Auto Loader, `.option("cloudFiles.schemaEvolutionMode", "addNewColumns")`, in combination with `.option("mergeSchema", "true")` together perform schema evolution. Auto Loader will handle changes in the schema for us. Auto Loader tracking the changes is beneficial when new data fields become available with or without prior notice; therefore, code changes are unnecessary.

There is documentation on all schema options. The method we use for the Streaming Transactions project is the only option for automatic schema evolution. Every other option will require manual intervention. For example, you can use the "`rescue`" mode to rescue data from being lost. Alternatively, you can use the "`failOnNewColumns`" mode to cause the pipeline to fail and keep the schema unchanged until the production code is updated. There are numerous options and patterns for Auto Loader. Check out *Common loading patterns with Auto Loader* docs linked in Further reading for more information.

Delta Live Tables for the Bronze layer

Delta Live Tables (**DLT**) is a Databricks feature that actively simplifies your pipeline operations, empowering you to focus on setting clear objectives for your pipeline rather than getting bogged down in operational details. Auto Loader's schema evolution capabilities are integrated with DLT, underscoring DLT's utility in handling dynamic data schemas with minimal manual intervention.